Chapter 12: Architecture Development Method

Study Guide – based on Mastering the TOGAF® Standard by Eric Jager (2025), Chapter 12 (pp. 295-560)

Overview

Chapter 12 (pp. 295-560) of *Mastering the TOGAF® Standard* by Eric Jager is the book's longest chapter. It covers the **Architecture Development Method (ADM)**, which the book calls “the heart of the TOGAF Standard.” For each ADM phase, the chapter gives the key objectives, a table of steps with their outputs, a practical walk-through of each step, and the artifacts that phase can produce, illustrated with the fictional healthcare organization from Chapter 5. It is part of the PINK, GREEN, and BLUE reading tours.

The chapter's most repeated message is that the **ADM describes what you *can* do, not what you *must* do**. Steps may be skipped or adapted, and the order is not fixed. TOGAF explicitly says the ADM must be **tailored**. It may look like a waterfall, but it is iterative (pp. 295-296, 547).

Study tip: Chapter 12 is mostly a long catalogue of artifacts with examples. For exams, focus on the phase objectives, the step tables, the four architecture documents, gap analysis, Transition Architectures, and the roadmap concepts. Know a few signature artifacts per phase, and use Chapter 7's Table 7-4 summary as your artifact checklist.

Detailed Section Breakdown

12.1 The Heart of the TOGAF Standard (pp. 296-300)

The ADM is often mistaken for a process. It is better seen as a set of tools for creating a sound EA. It has **ten phases** (Figure 12-1, p. 297), each with its own objectives:

Phase	Key objective(s)
Preliminary	Two objectives: (1) **determine** the Architecture Capability the organization needs, and (2) **establish** it in that form.
A – Architecture Vision	Establish a high-level vision and ensure stakeholder alignment. Business requirements are articulated and agreed.
B, C, D – Business / Information Systems / Technology	Two objectives: (1) craft the **Target Architecture** for the domain, and (2) identify **candidate Architecture Roadmap components** from the gaps between Baseline and Target.
E – Opportunities and Solutions	Three objectives: (1) identify the **work packages** needed to realize the Target, (2) decide whether an **incremental approach** (Transition Architectures) is required, and (3) define the **building blocks** needed to complete the Target.
F – Migration Planning	Create a **detailed plan** for implementing the architecture.
G – Implementation Governance	Ensure adherence to the architecture, manage changes during implementation, and continually assess progress.
H – Architecture Change Management	Continuously assess changes in strategy, business environment, and technology that may affect the architecture.
Requirements Management	Not a separate phase in practice. A **continuous** activity across the whole ADM that manages requirements in every phase.

Each phase has the same fixed components: **key objectives, steps, outputs, and approach**. Artifacts visualize the architecture, but they must **add value**. Don't produce them for their own sake (pp. 299-300).

12.2 Architecture Document Creation (pp. 300-305)

Four documents frame the architecture work. They record agreements and phases, not architecture detail:

Document	What it is	Created / used
Request for Architecture Work	A formal, high-level document from the **sponsoring organization** to the architecture team that triggers an ADM cycle.	Preliminary Phase. A new one is also created for major changes from Phase H or implementation.
Statement of Architecture Work	Defines the **scope and approach**. It is effectively a **contract between the architecting organization and the sponsor**. It includes the project description and scope, a vision overview, acceptance criteria and procedures, and optionally a plan, schedule, and approvals.	Deliverable of Phase A. Updated in B-D.
Architecture Definition Document	The **deliverable container** for the core artifacts. Spans all four domains and all states (**Baseline, Transition, Target**).	Built in B-D. Approved for the Target in Phase E. Finalized with Transition Architectures in Phase F.
Architecture Contract	Joint agreement between development partners and sponsors on **deliverables, quality, and fitness for purpose**. Depends on governance.	Created after the solution is chosen (E/F). Used in G and H.

Flow (Figure 12-2, p. 302): the Preliminary Phase produces the Request for Architecture Work. Phase A sketches the candidate building blocks and produces the Statement of Architecture Work. Phases B-D first describe the **Baseline** for each domain, reusing repository definitions, and then the **Target**, which is harder because it does not exist yet. Building blocks are documented with their rationale and validated against the repository, which favours reuse. **Gap analysis** then determines what to reuse, build, or remove, and the **Architecture Roadmap** is drafted. After formal stakeholder reviews, Phase E approves the document for the Target and links missing building blocks to work packages. The chosen solution goes into an Architecture Contract, which starts the transition and governance. Major changes later need a new Request for Architecture Work and a new ADM cycle (pp. 302-305).

12.3 Preliminary Phase (pp. 305-322)

This phase gets the organization ready for architecture by aligning stakeholders, defining principles, and tailoring TOGAF. Skipping it risks misalignment with business goals. It also determines current and target **architecture maturity** and **readiness for change** (p. 305).

Step (Tables 12-1 and 12-2)	Output
Define the enterprise: scope the impacted organization and identify stakeholders (the most affected, those whose work changes, and external parties)	Scope of impacted organization(s)
Identify key drivers and organizational context: scan strategies and business plans for drivers, goals, and objectives, then verify them with stakeholders	Baseline of objectives, goals, and drivers
Manage relationships between frameworks: reuse existing frameworks critically; set up the **Architecture Governance Framework** (repository requirements, compliance, retirement)	EA Capability structure; Architecture Governance Framework
Assess EA maturity: ACMM or CMMI (Ch. 15); Business Transformation Readiness Assessment (Ch. 14.7)	Maturity assessments, gaps, and strategies
Establish the EA team: the team need not be all architects; composition follows the capability's positioning; roles clarified with a RACI matrix (Table 12-3)	Architecture team; roles and responsibilities
Set Architecture Principles (Ch. 14.2.4)	Architecture Principles; constraints

Step (Tables 12-1 and 12-2)	Output
Choose and tailor the architecture framework: agreed glossary, tailored Enterprise Metamodel, tailored ADM (remove tasks, add checkpoints), tailored Content Framework, tools strategy, formal change control	Tailored framework, ADM, and Content Framework

- **Jager's key point:** *decisions in architecture are not the architect's responsibility, but the stakeholders'.* Stakeholders decide based on the architect's input (p. 311).
- **Hospital anecdote:** he distilled drivers, goals, and objectives from stacks of strategy documents into a **Driver/Goal/Objective Diagram** (Figure 12-3, p. 309), which over time grew into a Solution Concept Diagram. AI tools are now being used for this kind of document scanning (pp. 309-310).
- Jager calls CMMI the more appropriate maturity model for EA because it emphasizes business goals (p. 313).
- An “architecture team” is not the same as a team of people with the job title architect. Anyone involved in architecture work can be part of it (p. 315).
- **Artifact:** the **Principles Catalog** (entity: principle), which holds Enterprise and Architecture Principles. Each principle has a name, statement, rationale, and implications (Tables 12-4 and 12-5, pp. 320-323).

12.4 Phase A: Architecture Vision (pp. 324-347)

Phase A is the **first actionable phase**. It produces a high-level vision, stakeholder alignment, and initial approval to proceed. Inputs from the Preliminary Phase populate the first version of the Architecture Repository, for example capabilities stored as Architecture Building Blocks and the RACI matrix stored in the Architecture Capability section (p. 324).

Step (Table 12-6)	Output
Establish the architecture project: formal recognition, leadership endorsement, line management support; manage it like any project	Statement of Architecture Work
Identify stakeholders and determine requirements (Ch. 14.3)	Stakeholder Catalog; Communications Plan
Assess readiness for business transformation: factors such as vision, leadership, stakeholder engagement, governance, culture, and communications (Ch. 14.7)	Business Transformation Readiness Assessment
Elaborate on business goals and Architecture Principles; outline constraints; obtain senior management approval	Validated objectives, goals, drivers, and principles
Create a high-level architecture vision, including a business case	Solution Concept Diagram
Define Target Architecture value propositions and KPIs; high-level business risk assessment	Value propositions and KPIs
Identify business transformation risks and mitigation activities (Ch. 14.8)	Business transformation risks; mitigating activities
Develop the Statement of Architecture Work and secure formal approval (funding, resources, time)	Statement of Architecture Work

Phase A artifacts (Table 12-7, pp. 336-347)

Artifact	Purpose / notes
Stakeholder Catalog	Stakeholders, key concerns, classification (e.g., keep satisfied, key player, keep informed, minimal effort), and relevant artifacts. TOGAF uses **no metamodel entity** here because of sensitivity. Jager recommends adding a **stakeholder entity** when tailoring.
Value Chain Diagram	Primary and support activities (hospital example, Figure 12-4). Emphasizes presentational impact to engage stakeholders quickly.
Solution Concept Diagram	A "pencil sketch" linking objectives to stakeholder requirements and constraints. Despite the name, **not** a detailed solution.
Business Model Diagram	Also called the Business Model Canvas. Shows how value is created, delivered, and captured. Uses entities not in the Foundation metamodel (business collaboration, interaction, interface, resource, value).
Business Capability Map	High-level, goal-related capabilities. Jager notes it is really a **diagram**, since a map is textual (Ch. 7). It is refined in Phase B.
Value Stream Map	End-to-end value-adding activities. Again, really a diagram.

12.5 Phase B: Business Architecture (pp. 347-397)

Business Architecture should ideally be the **first architecture activity**. It lays the groundwork for all other domains and shows business value and return on investment to stakeholders. Strategy defines goals, drivers, and success metrics but not the path, and Business Architecture provides the path (pp. 347-348).

The common B/C/D step pattern (Table 12-9, pp. 350-358)

Phases B, C, and D follow the **same eight steps**. Only the domain changes:

#	Step	Output
1	**Select reference models, viewpoints, and tools** from the Architecture Repository	Views and viewpoints; required building blocks; architecture requirements
2	**Develop Baseline and Target descriptions**: only as much as is needed to support the vision; reuse existing descriptions; explore Target options with Architecture Alternatives and Trade-Offs (Ch. 14.9)	Baseline and Target Architecture descriptions
3	**Perform gap analysis** (Ch. 14.5): check consistency, validate against principles, objectives, and constraints, and compare Baseline with Target. Out-of-scope requirements go to the Requirements Repository.	Gap analysis results
4	**Define candidate roadmap components**: gaps become work packages (initiatives), forming a draft roadmap that is refined in Phase E	Draft Architecture Roadmap components
5	**Resolve impacts across the Architecture Landscape**: effects on existing architectures, recent changes, reuse opportunities, and interactions with other projects	Validated building blocks and artifacts
6	**Conduct formal stakeholder review** with executives, process owners, and SMEs against the original Statement of Architecture Work	(Updated) Statement of Architecture Work
7	**Finalize the architecture**: standardize and document building blocks, record the rationale for decisions, link requirements to elements	Documented building blocks and requirements
8	**Create or update the Architecture Definition Document**, at minimum covering Baseline and Target	Architecture Definition Document

Business Architecture techniques (pp. 359-365)

- **Applying business capabilities:** link the Phase A capability map to organization units, value streams, information systems, and objectives.
- **Business Capability Heat Mapping** (Figure 12-9, p. 361): colours show maturity, performance, value, or cost. Example scheme: **green** = at target, **yellow** = one level below, **red** = two or more levels below, **purple** = desired but not yet in place. Jager: there is no scientific calculation; it relies on stakeholder alignment, gut feeling, and common sense.
- **Value streams** give the *why*; capabilities give the *what*. Mapping value stream stages to capabilities enables gap analysis.
- **Organization Map:** a web of working relationships among units, partners, and stakeholders, not a reporting chart. Shows who owns capabilities and takes part in value streams.
- **Information Maps:** the core information concepts of the business, cross-mapped to capabilities.
- **Modeling techniques:** activity models (BPMN from the Object Management Group), use-case models, logical data or class models (is-a relationships).
- **Business Scenarios:** used mainly in the Preliminary, A, and B phases to test the assumptions behind a project.

Phase B artifacts (Table 12-10, pp. 366-397)

Phase B has the most artifacts. Catalogs include Organization/Actor, Driver/Goal/Objective, Role, Business Service/Function, Location, Process/Event/Control/Product, Contract/Measure, Business Capabilities, Value Stream (Stages), and Business Glossary. Matrices include Value Stream/Capability, Strategy/Capability, Capability/Organization, Business Interaction, and Actor/Role. Diagrams include Business Footprint, Business Service/Information, Functional Decomposition, Product Lifecycle, Goal/Objective/Business Service, Business Use-Case, Organization Decomposition, Process Flow, and Business Event. Maps include the Business Capability Map, Value Stream Map, Organization Map, and Information Map. Jager's notable comments: the Business Service/Function **Catalog** works better as a **matrix** (Table 12-15), and several “maps” are really diagrams.

12.6 Phase C: Information Systems Architectures (pp. 398-441)

Phase C develops the blueprint of the information systems that support the Business Architecture. It is split into **Data Architecture** (data assets, management, and governance) and **Application Architecture** (applications, their interactions, and users). Its objectives and steps mirror Phase B (Tables 12-31 and 12-38). Jager notes that BIZBOK's **information concept** sits between Business and Data Architecture but is not yet reflected in TOGAF's Fundamental Content. Only the *Information Mapping* Series Guide covers it (pp. 398-399).

Sub-phase	Artifacts
Data Architecture (Table 12-32, p. 411)	Data Entity/Data Component Catalog; Data Entity/Business Function Matrix; Application/Data Matrix; Conceptual Data Diagram; Data Dissemination Diagram; Data Security Diagram; Data Migration Diagram; Data Lifecycle Diagram
Application Architecture (Table 12-39, p. 428)	Application Portfolio Catalog (example uses a CIA score: confidentiality, integrity, availability; Figure 12-31); Interface Catalog; Application/Organization, Role/Application, Application/Function, and Application Interaction Matrices; Application Communication, Application and User Location, Application Use-Case, Enterprise Manageability, Process/Application Realization, Software Engineering, Application Migration, and Software Distribution Diagrams

12.7 Phase D: Technology Architecture (pp. 441-459)

Phase D defines the hardware, software, network, and platform services, and the **technology standards**, needed to support the Business, Data, and Application Architectures, with a focus on consistency, scalability, security, and interoperability. It uses the same steps as B and C (Table 12-46). **Jager:** in nine out of ten organizations, IT plans separately. TOGAF's idea is to link technology work back to business goals (p. 442).

Artifacts (Table 12-47, p. 453): Technology Standards Catalog; Technology Portfolio Catalog; Application/Technology Matrix; Environments and Locations Diagram; Platform Decomposition Diagram; Processing Diagram; Networked Computing/Hardware Diagram; Network and Communications Diagram.

12.8 Phase E: Opportunities and Solutions (pp. 459-495)

Phase E turns the architecture into **projects and programs** (work packages). It evaluates approaches such as **buy, build, or reuse**, and creates the **first complete Architecture Roadmap** from the gap analyses and candidate components of B-D. It also refines Architecture Building Blocks into Solution Building Blocks (pp. 459-460). Jager describes it as the bridge between defining the architecture and planning its implementation.

Step (Table 12-52)	Output
Determine/confirm key corporate change attributes: culture, skills, and the transition capabilities of units	Implementation Factor Catalog (risks, issues, assumptions, dependencies, actions, impacts)
Determine business constraints for implementation; mediate using Alternatives and Trade-Offs	—
Review and consolidate gap analysis results from B-D: one list of gaps, solutions, and dependencies, using cross-domain views to find dependencies	Consolidated Gaps, Solutions, and Dependencies Matrix
Consolidate and reconcile interoperability requirements: resolve conflicts by building a translating building block or by changing specifications	—
Refine and validate dependencies: group related activities into increments	Work Package Portfolio Map
Confirm readiness and risk for business transformation: revisit the Phase A assessment	Updated Consolidated Matrix
Formulate the Implementation and Migration Strategy	Implementation and Migration Strategy
Identify and group major work packages: classify current systems as **mainstream, contain, or replace**	Work Package Portfolio Map
Identify Transition Architecture(s): milestones that each deliver business value; intervals need not be equal; schedule hard activities after easier ones that deliver missing capabilities	Architecture Definition Increments Table; Transition Architectures
Create the Architecture Roadmap and the Implementation and Migration Plan (Table 12-53)	Architecture Roadmap; Implementation and Migration Plan

Strategy choice	Options (p. 470)
Basic implementation approach	**Greenfield** (completely new), **Revolutionary** (radical change, e.g., full overhaul), **Evolutionary** (gradual convergence, e.g., parallel running or phased introduction)
Implementation methodology	**Quick Wins**, **Achievable Targets**, **Value Chain Method**

- **Jager's artifact nuance:** a catalog or matrix only counts as an *artifact* if it contains architectural **entities**. The Implementation Factor Catalog and the Consolidated Gaps Matrix contain none, so strictly they are not artifacts (pp. 463-464).
- **Phase E artifacts (Table 12-54):** Project Context Diagram (the only TOGAF use of the work package entity), Benefits Diagram, Implementation Factor Catalog, Consolidated Gaps, Solutions, and Dependencies Matrix,

Architecture Definition Increments Table, and, as Jager's additions, the **Work Package Portfolio Map** and the **Architecture Roadmap**.

- **Plateau Planning** (Figures 12-42 and 12-43) visualizes Baseline, Transition, and Target plateaus. It can show projects instead of gaps.
- A data-driven sequence for implementation (implement applications that create data before those that process it) is one valid planning approach (p. 474).

12.8.3 Creating an Architecture Roadmap (pp. 485-494)

- The roadmap guides the move from Baseline through Transition to Target (Figure 12-44). It is built incrementally in E and F from the components identified in B-D, and it also works as a communication tool.
- **Work package (TOGAF definition):** *a set of actions identified to achieve one or more objectives for the business; it can be part of a project, a complete project, or a program*. It is like a process, but performed **once** with a defined end result (Figure 12-45). In agile, a work package can model a sprint or an increment.
- Clusters of work packages form initiatives, projects, programs, or portfolios.
- **Work Package Portfolio Map** (Table 12-59): work packages linked to objectives and goals. It is the foundation of the roadmap. In Phase F, add a projects column to avoid duplicate work.
- **Rule:** *a roadmap should never be created without linking business goals and objectives to the initiatives.* Goals and objectives must be **SMART** (Specific, Measurable, Achievable, Realistic, Timebound).
- Roadmap layout: goals and objectives run down the left, and work packages are plotted over time with attributes such as status, owner, and cost (Figures 12-47 and 12-48).

12.9 Phase F: Migration Planning (pp. 495-513)

Phase F produces a detailed implementation plan. It prioritizes projects, assesses dependencies and risks, plans resources, and ensures business continuity. The Architecture Roadmap is **finalized** here (p. 495).

Step (Table 12-60)	Output
Confirm management framework interactions: align with **Business Planning, Enterprise Architecture, Project/Portfolio Management, and Operations Management**	Updated Implementation and Migration Plan
Assign business value to work packages: identify projects and assess risks	Updated Work Package Portfolio Map; Business Value and Risk Matrix
Estimate resources, timings, and delivery vehicle: separate capital from operational expenses; find savings	—
Prioritize migration projects through cost/benefit assessment and risk validation	**Risk Management Plan** (at minimum: document information, approvals, approach)
Confirm the Architecture Roadmap with any Transition Architectures	Transition Architecture State Evolution Table; finalized roadmap
Complete the Implementation and Migration Plan with milestones per transition	Finalized Implementation and Migration Plan
Complete the architecture development cycle: an Implementation Governance Model if maturity allows; lessons learned (feed into Phase H)	Implementation Governance Model; lessons learned

- **Transition Architecture State Evolution Table** (Figure 12-49): rows of domain and service, columns of Transition Architectures, and each cell describes the planned state (in transition, to be removed, new or retained). It is demanding to build, but worth it.
- **Business Value and Risk Matrix** (Figure 12-50): business value index (compliance, strategic alignment, competitive position) on the y-axis against a risk index (size and complexity, technology, organizational capacity, impact of failure) on the x-axis. Criteria are weighted and **approved by senior management before plotting**. Colours flag risk.
- If the roadmap needs more detail than the Architecture Definition Document provides, that signals a move to a different level of architecture (Phase F).

12.10 Phase G: Implementation Governance (pp. 513-528)

Phase G is where the vision becomes operational reality. It applies the **Chapter 8 governance framework** to make sure implementation projects follow the architecture (pp. 513-514).

Step (Table 12-62)	Output
Confirm scope and priorities: gap analysis on deployment, Solution Building Blocks led by Solution Architects, reuse of past investments, workshop with development management	Architecture Contract
Identify deployment resources and skills: development methods, modeling tools, team readiness, dependencies	—
Guide development of solutions deployment: project recommendations, strategic requirements	Change requests
Perform architecture compliance reviews	Compliance assessments
Implement business and IT operations: service delivery, skills development, training	Architecture-compliant solutions
Perform post-implementation review: deviations lead to change requests	Publication of reviews

12.10.2 Architecture performance monitoring (pp. 522-528)

- Two things to measure: **completion of work packages** and **achievement of goals and objectives**.
- **Basic method:** split a work package evenly across its activities (four activities = 25% each). **Refined method:** weight by activity duration (e.g., 12.5 / 25 / 25 / 37.5%). With three of four activities done, the basic method gives **75%** and the refined method gives **62.5%**, which is more realistic (Figures 12-51 and 12-52).
- Visualize with pie charts per work package (Table 12-63), or spider or bar charts. The choice is personal preference.
- **Goal progress** = the average of its work packages' percentages. Example: Goal A (WP A 100%, WP B 75%) = **87.5%**. Goal B (WP C 50%, D 0%, E 0%) = **16.7%** (Figures 12-53 and 12-54). This links to the Chapter 8 KPI dashboards.

12.11 Phase H: Architecture Change Management (pp. 528-539)

Architecture is not static. Phase H keeps it aligned as strategy, the business environment, and technology change. It assesses impact, defines change procedures, implements changes with minimal disruption, and reviews afterwards (p. 528).

Step (Table 12-64)	Output
Establish a value realization process: governs when and how the EA may change and what triggers a new ADM cycle; linked to the governance framework and the Architecture Contract	Architecture Compliance Report
Deploy monitoring tools: for business and technology change and for the capability's maturity	—
Manage risks before business transformation	Risk Management Plan
Provide analysis for architecture change management: monitoring data, feedback, performance reviews	—
Develop change requirements to meet performance targets	**Minor changes**: changes to the framework or principles (possibly updating the metamodel or Content Framework). **Major changes**: a **new Request for Architecture Work**
Manage the governance process: policies for proposing, reviewing, and approving changes	Architecture Contract
Activate the process to implement change: rollout plan with timelines, dependencies, and milestones	—

Jager warns that organizations are often left with an EA that fit *yesterday's* organization, or one where the architecture still fits but the underlying solutions do not (p. 534).

12.12 Requirements Management (pp. 539-547)

Requirements Management is a **continuous activity** across all phases. It captures, tracks, categorizes, prioritizes, versions, and refines requirements so the architecture stays aligned with business goals. Its key objective is to make sure the requirements management process is **available and applied in every phase** (pp. 539-541).

#	Step (Table 12-65)	Who
1	Identify and document requirements (from Phase A, into the Requirements Repository; Figure 12-55)	ADM phase
2	Establish baseline requirements in the Requirements Repository (B-D, prioritized in E)	Req. Mgmt
3	Monitor baseline requirements	Req. Mgmt
4	Identify new and changed requirements	ADM phase
5	Identify changed requirements and record priorities (validate with stakeholders; manage conflicts)	Req. Mgmt
6	Assess the impact of changed requirements on current and previous phases (implement now or in a later iteration)	ADM phase
7	Implement requirements arising from Phase H	ADM phase
8	Update the Architecture Requirements Repository	Req. Mgmt
9	Implement change in the current phase (no Req. Mgmt action)	ADM phase
10	Assess and revise gap analysis for past phases (**gap requirements**)	ADM phase

Artifact: a single essential deliverable, the **Requirements Catalog** (entities: requirement, objective, goal, assumption, constraint, gap), often accompanied by a requirements management plan. Link each requirement to a goal or objective for traceability (Tables 12-66 and 12-67, pp. 546-547).

12.13 The ADM in Short and 12.14 Summary (pp. 547-560)

Section 12.13 recaps every phase's steps in one list, with a poster in Figure 12-56 (p. 557). Jager repeats that the ADM is a **logical method grouping key activities**, not a process to execute in full. The most common misconception is that *everything* must be done. Steps can be skipped (especially in agile), the order is not fixed, and the ADM is iterative (Ch. 13.3).

Key Terms & Definitions

Architecture Development Method (ADM) (p. 295) — TOGAF's iterative, tailorable method for developing and managing EA. Ten phases, including Requirements Management.

Request for Architecture Work (p. 300) — Sponsor-to-architecture-team document that triggers an ADM cycle.

Statement of Architecture Work (p. 301) — Scope and approach of the cycle. Effectively a contract between the architects and the sponsor (Phase A).

Architecture Definition Document (p. 301) — Deliverable container for the core artifacts across all domains and states.

Architecture Contract (p. 301) — Agreement between development partners and sponsors on deliverables, quality, and fitness for purpose.

Baseline / Transition / Target Architecture (pp. 302-304) — Current state, intermediate value-delivering states, and desired future state.

Gap analysis (pp. 304, 353) — Comparing Baseline and Target to decide what to reuse, build, or remove (Ch. 14.5).

Architecture Principles vs. Enterprise Principles (p. 320) — Enterprise Principles guide overall strategy, culture, and decisions. Architecture Principles describe what a good architecture or solution looks like.

Business Transformation Readiness Assessment (p. 329) — Evaluates and quantifies readiness for change (vision, leadership, engagement, governance, culture).

Solution Concept Diagram (p. 343) — A high-level “pencil sketch” linking objectives, requirements, and constraints.

Business Capability Heat Map (p. 361) — A colour-coded capability map showing maturity, performance, value, or cost.

Implementation Factor Catalog (p. 462) — Records the factors that influence the Implementation and Migration Plan.

Consolidated Gaps, Solutions, and Dependencies Matrix (p. 465) — One list of B-D gaps with their potential solutions and dependencies.

Greenfield / Revolutionary / Evolutionary (p. 470) — Basic approaches in the Implementation and Migration Strategy.

Transition Architecture (p. 472) — An intermediate architecture state that delivers measurable business value on the way to the Target.

Work package (p. 487) — A set of actions to achieve one or more business objectives. Can be part of a project, a project, or a program.

Work Package Portfolio Map (p. 488) — Jager's added artifact linking work packages to objectives and goals. The foundation of the roadmap.

Architecture Roadmap (p. 485) — Work packages plotted over time from Baseline to Target, linked to goals and objectives.

Business Value and Risk Matrix (p. 511) — Projects plotted on weighted business-value and risk indices (Phase F).

Minor vs. major change (p. 534) — Minor: update the framework or principles. Major: a new Request for Architecture Work and a new ADM cycle.

Requirements Catalog (p. 546) — The single essential Requirements Management artifact.

Figures & Tables Referenced (key items)

Item	Page	Why it matters
Figure 12-1 – The Architecture Development Method	p. 297	The ADM cycle itself.
Figure 12-2 – Document creation during the ADM	p. 302	Where the four architecture documents appear.
Tables 12-1 / 12-2 – Preliminary Phase steps	pp. 306, 314	Determine and establish the capability.
Table 12-3 – Example RACI for an EA team	p. 315	Links to Ch. 8's RACI.
Table 12-5 – Principles Catalog	p. 321	Name, statement, rationale, implications.
Table 12-6 – Phase A steps	p. 325	Vision, stakeholders, readiness, Statement of Architecture Work.
Table 12-8 – Stakeholder Catalog	p. 338	Concerns, classification, relevant artifacts.
Table 12-9 – Phase B steps	p. 350	The step template reused for C and D.
Figure 12-9 – Business Capability Heat Map	p. 361	Colour-coded capability assessment.
Tables 12-10 / 12-32 / 12-39 / 12-47 – Artifacts for B, C-Data, C-Application, D	pp. 367, 411, 428, 453	Artifact checklists per domain.
Table 12-52 – Phase E steps	p. 461	Ten steps from change attributes to roadmap.
Table 12-57 – Consolidated Gaps, Solutions, and Dependencies Matrix	p. 483	Gap, solution, and dependency example.
Figure 12-42 – Plateau Planning	p. 482	Baseline, Transition, and Target plateaus.
Figure 12-44 – From Baseline through Transition to Target	p. 486	The roadmap concept.
Table 12-59 – Work Package Portfolio Map	p. 489	Work packages linked to objectives and goals.
Table 12-60 – Phase F steps	p. 496	Migration planning.
Figure 12-50 – Business Value and Risk Matrix	p. 512	Prioritization tool.
Table 12-62 – Phase G steps	p. 514	Implementation governance.
Figures 12-51 to 12-54 – Progress and goal realization	pp. 523-527	Monitoring calculations.
Table 12-64 – Phase H steps	p. 529	Change management; minor vs. major change.
Table 12-65 – Requirements Management steps	p. 541	The ten steps.
Figure 12-56 – Summary of ADM steps	p. 557	One-page poster of the whole ADM.

Key Takeaways / Exam Focus Points

- The ADM is the **heart** of TOGAF. It has **ten phases** (Preliminary, A-H, Requirements Management). It is iterative, tailorable, and non-prescriptive: steps are optional and their order is not fixed (pp. 295-296, 547).
- Every phase has the same four components: objectives, steps, outputs, approach (p. 300).
- The four documents and when they appear: **Request for Architecture Work** (Preliminary, from the sponsor) → **Statement of Architecture Work** (Phase A, a contract with the sponsor) → **Architecture Definition Document** (B-D, approved in E, finalized in F) → **Architecture Contract** (E/F onward, used in G and H) (pp. 300-305).
- Preliminary has two objectives: **determine** and **establish** the Architecture Capability. Its artifact is the Principles Catalog (pp. 305, 320).
- Phase A is the first actionable phase. Key outputs: the Statement of Architecture Work, Stakeholder Catalog, Communications Plan, Business Transformation Readiness Assessment, and Solution Concept Diagram (p. 325).
- B, C, and D share one step pattern: reference models → Baseline and Target → **gap analysis** → candidate roadmap components → resolve impacts → stakeholder review → finalize → Architecture Definition Document (pp. 350, 400, 419, 443).
- Business Architecture comes first because the other domains build on it (p. 348).
- Phase E: work packages, the incremental decision (Transition Architectures), and building blocks. Strategy approaches are **Greenfield, Revolutionary, Evolutionary**. Methodologies are **Quick Wins, Achievable Targets, Value Chain Method**. Systems are classified as **mainstream, contain, or replace** (pp. 460, 470-471).
- Every roadmap initiative must link to SMART goals and objectives (p. 491).
- Phase F: the four management frameworks (Business Planning, EA, Project/Portfolio Management, Operations Management), the Business Value and Risk Matrix, the Risk Management Plan, and the finalized roadmap (pp. 496-512).
- Phase G applies the Chapter 8 governance framework: compliance reviews, Architecture Contract, post-implementation review, and progress monitoring (75% basic vs. 62.5% refined; goal average 87.5% / 16.7%) (pp. 514-527).
- Phase H: minor changes update the framework or principles; **major changes require a new Request for Architecture Work** (p. 534).
- Requirements Management is continuous. Its single artifact is the Requirements Catalog (pp. 540, 546).
- Jager's additions and critiques: add a stakeholder entity; maps that are really diagrams; matrices preferred over some catalogs; the Work Package Portfolio Map and Architecture Roadmap as new artifacts; catalogs without entities aren't true artifacts.

Self-Check Questions

Q: How many phases does the ADM have, and which is not really a separate phase?

A: Ten: Preliminary, A through H, and Requirements Management. Requirements Management is a continuous activity across all phases (pp. 297-299).

Q: Does TOGAF require all ADM steps to be performed in order?

A: No. The ADM describes what can be done. Steps may be skipped or adapted, the order is not fixed, and the ADM must be tailored (pp. 295-296, 547).

Q: Name the four architecture documents and the phase where each originates.

A: Request for Architecture Work (Preliminary, from the sponsor); Statement of Architecture Work (Phase A); Architecture Definition Document (Phases B-D, approved in E, finalized in F); Architecture Contract (after the solution is chosen, used in G and H) (pp. 300-305).

Q: What are the two objectives of the Preliminary Phase?

A: To determine the Architecture Capability the organization needs and to establish it in that form (p. 305).

Q: What is the difference between Enterprise Principles and Architecture Principles?

A: Enterprise Principles are high-level guidelines shaping overall strategy, culture, and decision-making. Architecture Principles describe what a good solution or architecture looks like (p. 320).

Q: Why does TOGAF use no metamodel entity for the Stakeholder Catalog, and what does Jager recommend?

A: Because stakeholder mapping is potentially sensitive and Phase A uses informal modeling. Jager recommends adding a stakeholder entity when tailoring the metamodel (pp. 336-338).

Q: What are the two shared objectives of Phases B, C, and D?

A: To craft the Target Architecture for the domain, and to identify candidate roadmap components from the gaps between Baseline and Target (p. 348).

Q: In a maturity heat map, what do green, yellow, red, and purple mean?

A: Green: at the desired maturity. Yellow: one level below. Red: two or more levels below. Purple: desired but not yet in place (or surplus resources) (p. 361).

Q: Name the three basic implementation approaches and three implementation methodologies in Phase E.

A: Approaches: Greenfield, Revolutionary, Evolutionary. Methodologies: Quick Wins, Achievable Targets, Value Chain Method (p. 470).

Q: What is a Transition Architecture, and must transitions be equally spaced?

A: An intermediate architecture state that delivers measurable business value on the way to the Target. The intervals need not be uniform (p. 472).

Q: What is TOGAF's definition of a work package, and how does it differ from a process?

A: A set of actions identified to achieve one or more business objectives, which can be part of a project, a project, or a program. Unlike a process, it is performed once and produces a defined end result (pp. 486-487).

Q: Which four management frameworks must Phase F coordinate with?

A: Business Planning, Enterprise Architecture, Project/Portfolio Management, and Operations Management (pp. 497-498).

Q: What two axes form the Business Value and Risk Matrix, and who approves its criteria?

A: A business value index (compliance, strategic alignment, competitive position) and a risk index (size and complexity, technology, organizational capacity, impact of failure). Senior management approves the criteria and weights before plotting (p. 511).

Q: A work package has activities lasting 1, 2, 2, and 3 weeks, and the first three are done. What progress do the basic and refined methods show?

A: Basic: 75% (25% each). Refined (duration-weighted 12.5/25/25/37.5): 62.5% (p. 524).

Q: In Phase H, what distinguishes a minor change from a major change?

A: Minor changes adjust the architecture framework or principles, possibly updating the metamodel or Content Framework. Major changes require a new Request for Architecture Work and start a new ADM cycle (p. 534).

Q: What is the single essential artifact of Requirements Management, and which entities does it use?

A: The Requirements Catalog. Entities: requirement, objective, goal, assumption, constraint, gap (p. 546).